Synthetic Banking Mock Data Design and Delivery Guide

Confluence-ready reference for the synthetic Bronze/source environment

Context

Transaction investigation context: mock transactions, dispute cases, merchant categories, and investigation notes.

Repository: mock-data-generator

This document describes a synthetic banking Bronze/source environment used to test transaction investigation, data engineering, data-quality controls, and downstream analytics. It is intentionally self-contained and uses safe masked examples only; it does not disclose real customer values, confidential source details, or internal locations.

Scope at a glance

Item	Design
Published scope	41 source tables across four domains
Delivery series	Six complete daily snapshots: 5–10 July 2026
Latest full snapshot	6,241,418 rows on 10 July 2026
Data model	Valid relational base population followed by deterministic, labelled error injections
Raw-data boundary	Synthetic PII, payment identifiers, and investigations are restricted; raw Bronze and the truth manifest are not AI inputs

Generation and delivery flow

MERMAID

```
flowchart LR
    A[Configuration and deterministic seed] --> B[Generate valid Customer Master base]
    B --> C[Generate valid Customer Transaction and Card base]
    C --> D[Generate valid Financial Crime relationships]
    D --> E[Apply labelled error-injection rules]
    E --> F[Create local-only truth manifest]
    E --> G[Run quality and referential checks]
    G --> H[Reconstruct full daily snapshot]
    H --> I[Audit published tables and row counts]
    I --> J[Publish audited Parquet full snapshot to controlled S3]
    F -. never published / never direct AI access .-> K[Restricted evidence boundary]
    J --> L[Approved downstream data consumers]
    L --> M[Masked or tokenised AI-safe view]
```

The generator starts from an internal 4 July seed. For each 5–10 July business date it creates source-system changes, resolves the clean base dependency order, applies controlled defects, and reconstructs a full point-in-time snapshot. Only audited full Parquet snapshots are delivered; raw daily extracts, manifests, build state, and quality reports remain excluded.

Relationship model

MERMAID

```
flowchart TB
    CUST[Customer identities
core banking / CRM] --> HOLD[Customer-account holder bridge]
    HOLD --> ACCT[Account]
    CUST --> KYC[KYC, employment, service requests]
    ACCT --> ATXN[Account transaction]
    ACCT --> CARD[Card]
    CARD --> CTXN[Card transaction]
    ATXN --> AEV[Account status events]
    CTXN --> CEV[Card status events]
    ATXN --> RISK[Risk score and fraud alert]
    ATXN --> TM[Monitoring alert]
    CTXN --> TM
    RISK --> CASE[Investigation case]
    TM --> CASE
    CASE --> EVID[Case evidence bridges]
    CASE --> NOTE[Investigation notes]
    CASE --> AML[AML case]
    AML --> SAR[Suspicious-activity report]
    SCREEN[Sanctions screening / watchlist] --> CASE
    ATXN --> GATEWAY[Gateway and ATM operational logs]
    CTXN --> GATEWAY
```

The arrows show logical source relationships. A mixed customer reference in a raw table can point to either a core-banking or CRM source identifier; identity conformance occurs downstream, not in Bronze.

Domain volumes

Domain	Tables	10 July rows	Primary purpose
Customer Master	7	560,613	Identity, KYC, employment, servicing, accounts, and holders
Customer Transaction	10	3,240,965	Account postings, status events, merchant context, ATM/gateway logs, balances
Card	5	1,659,882	Card instruments, card transactions, lifecycle events, fraud flags, limits
Financial Crime	19	779,958	Risk scores, alerts, cases, AML/sanctions, evidence, notes, reporting

Domain	Tables	10 July rows	Primary purpose
Total	**41**	**6,241,418**	Full point-in-time source image

Source-table data dictionary

Reading the table dictionaries

- **Required** means yes when the raw schema declares a primary key or non-null field. A nullable value can still be made invalid by a labelled error rule.
- **Key / reference** distinguishes primary/unique keys from source references. A reference may intentionally be unresolved in an injected row.
- **Classification** tells consumers whether a field is internal, sensitive, restricted, contact, or financial/risk data.
- **Error injected** gives the exact controlled condition, rate, and expected downstream handling for the field or event row.
- **Injected example** shows the safe, non-production changed value; no real names, identifiers, phone numbers, email addresses, accounts, PANs, or confidential content are shown.

Customer Master

core_banking_customer

10 July volume: 87,369 rows

Grain: source entity / relationship snapshot (SCD2-ready)

Primary key: cust_no

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
cust_no	varchar	yes	PK	source-controlled / free text	internal	—	—
cif_number	varchar	no	Unique	source-controlled / free text	internal	—	—
full_name	varchar	no	—	source-controlled / free text	direct identifier	—	—
date_of_birth	date	no	—	ISO date	sensitive	Set to NULL. **Rate:** 2.5% **Handling:** QUARANTINE_FIELD	NULL

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
national_id	varchar	no	—	source-controlled / free text	sensitive	Copy another row's value, creating a duplicate national ID. **Rate:** 0.8333% **Handling:** DEDUPLICATE Set to NULL. **Rate:** 4.1667% **Handling:** QUARANTINE_FIELD	[copied synthetic identifier] NULL
phone	varchar	no	—	source-controlled / free text	contact	Replace with the placeholder 0 000 000 000. **Rate:** 2.5% **Handling:** QUARANTINE_FIELD	[placeholder synthetic phone]
address	varchar	no	—	source-controlled / free text	sensitive	—	—
source_system	varchar	no	—	source-controlled / free text	internal	—	—
created_date	date	no	—	ISO date	internal	—	—

crm_customer

10 July volume: 65,527 rows

Grain: source entity / relationship snapshot (SCD2-ready)

Primary key: party_id

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
party_id	varchar	yes	PK	source-controlled / free text	internal	—	—
customer_name	varchar	no	—	source-controlled / free text	direct identifier	—	—
phone	varchar	no	—	source-controlled / free text	contact	—	—
national_id	varchar	no	—	source-controlled / free text	sensitive	Set to NULL. **Rate:** 5.0% **Handling:** QUARANTINE_FIELD	NULL

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
email	varchar	no	—	source-controlled / free text	contact	Set to NULL. **Rate:** 3.125% **Handling:** QUARANTINE_FIELD Replace with malformed-email. **Rate:** 1.875% **Handling:** QUARANTINE_FIELD	NULL malformed-email
preferred_contact_method	varchar	no	—	source-controlled / free text	internal	—	—
source_system	varchar	no	—	source-controlled / free text	internal	—	—
created_date	date	no	—	ISO date	internal	—	—

customer_kyc

10 July volume: 78,636 rows

Grain: source entity / relationship snapshot (SCD2-ready)

Primary key: kyc_id

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
kyc_id	varchar	yes	PK	source-controlled / free text	internal	—	—
customer_ref	varchar	no	—	source-controlled / free text	internal	Replace with unresolved UNKNOWN-CUSTOMER. **Rate:** 1.0% **Handling:** QUARANTINE_RECORD	UNKNOWN-CUSTOMER
id_type	varchar	no	—	source-controlled / free text	internal	—	—
id_number	varchar	no	—	source-controlled / free text	sensitive	Replace with malformed INVALID-ID. **Rate:** 2.0% **Handling:** QUARANTINE_FIELD	INVALID-ID
verification_status	varchar	no	—	source-controlled / free text	internal	—	—
verified_date	date	no	—	ISO date	internal	—	—

customer_employment

10 July volume: 52,424 rows

Grain: source entity / relationship snapshot (SCD2-ready)

Primary key: employment_id

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
employment_id	varchar	yes	PK	source-controlled / free text	internal	—	—
customer_ref	varchar	no	—	source-controlled / free text	internal	Replace with unresolved UNKNOWN-CUSTOMER. **Rate:** 1.0% **Handling:** QUARANTINE_RECORD	UNKNOWN-CUSTOMER
employer_name	varchar	no	—	source-controlled / free text	direct identifier	—	—
job_title	varchar	no	—	source-controlled / free text	internal	—	—
monthly_income	decimal(12,2)	no	—	numeric range	financial / risk	Set to −1000.00, below the valid minimum of zero. **Rate:** 1.0% **Handling:** QUARANTINE_RECORD	−1000.00

customer_request

10 July volume: 38,139 rows

Grain: source entity / relationship snapshot (SCD2-ready)

Primary key: request_id

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
request_id	varchar	yes	PK	source-controlled / free text	internal	—	—
customer_ref	varchar	no	—	source-controlled / free text	internal	Replace with unresolved UNKNOWN-CUSTOMER. **Rate:** 1.0% **Handling:** QUARANTINE_RECORD	UNKNOWN-CUSTOMER
request_type	varchar	no	—	source-controlled / free text	internal	—	—

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
channel	varchar	no	—	source-controlled / free text	internal	—	—
request_date	date	no	—	ISO date	internal	—	—
status	varchar	no	—	source-controlled / free text	internal	—	—
resolution_date	date	no	—	ISO date	internal	—	—
description	varchar	no	—	source-controlled / free text	restricted content	Set to NULL. **Rate:** 4.0% **Handling:** QUARANTINE_FIELD	NULL

customer_account

10 July volume: 124,937 rows

Grain: source entity / relationship snapshot (SCD2-ready)

Primary key: link_id

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
link_id	varchar	yes	PK	source-controlled / free text	internal	—	—
cif_number	varchar	yes	—	source-controlled / free text	internal	—	—
account_id	bigint	yes	—	source-controlled / free text	internal	Replace with unresolved −1. **Rate:** 1.0% **Handling:** QUARANTINE_RECORD	−1
relationship_type	varchar	no	—	source-controlled / free text	internal	—	—
linked_date	date	no	—	ISO date	internal	—	—

account

10 July volume: 113,581 rows

Grain: source entity / relationship snapshot (SCD2-ready)

Primary key: account_id

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
account_id	bigint	yes	PK	source-controlled / free text	internal	—	—
customer_ref	varchar	no	FK → core_banking_customer.cust_no	source-controlled / free text	internal	—	—
cif_number	varchar	no	—	source-controlled / free text	internal	Replace with unresolved CIF99 999 999. **Rate:** 1.0% **Handling:** QUARANTINE_RECORD	CIF99 999 999
product_type	varchar	no	—	source-controlled / free text	internal	—	—
status	varchar	no	—	source-controlled / free text	internal	—	—
open_date	date	no	—	ISO date	internal	—	—
branch_code	varchar	no	—	source-controlled / free text	internal	—	—
servicing_model	varchar	no	—	source-controlled / free text	internal	—	—

Customer Transaction

account_transaction_status_event

10 July volume: 1,744,290 rows

Grain: immutable status-change event (CDC)

Primary key: status_event_id

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
status_event_id	varchar	yes	PK	source-controlled / free text	internal	Duplicate the status-event row as a replay. **Rate:** 1.0% **Handling:** DEDUPLICATE	[replayed immutable event row]
account_txn_id	bigint	yes	Unique	source-controlled / free text	internal	Replace with unresolved −1. **Rate:** 1.0% **Handling:** QUARANTINE_RECORD	−1
status	varchar	yes	—	INITIATED / PENDING / SUCCESS / REJECTED / CANCELED	internal	—	—

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
status_timestamp	timestamp	yes	—	ISO-8601 timestamp	internal	—	—
source_arrival_timestamp	timestamp	yes	—	ISO-8601 timestamp	internal	Shift the sequence-0 event 600 seconds later. Business time and sequence remain correct. **Rate:** 1.0% of initial events **Handling:** PROCESS_BY_ARRIVAL_ORDER	source-arrival timestamp +600 seconds
sequence_number	bigint	yes	—	source-controlled / free text	internal	—	—

atm_transaction_status_event

10 July volume: 264,330 rows

Grain: immutable status-change event (CDC)

Primary key: status_event_id

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
status_event_id	varchar	yes	PK	source-controlled / free text	internal	—	—
log_id	varchar	yes	FK → log_atm.log_id	source-controlled / free text	internal	Replace with unresolved UNKNOWN-ATM-LOG. **Rate:** 1.0% **Handling:** QUARANTINE_RECORD	UNKNOWN-ATM-LOG
account_txn_id	bigint	no	FK → account_transaction.account_txn_id	source-controlled / free text	internal	—	—
status	varchar	yes	—	source-controlled / free text	internal	—	—
status_timestamp	timestamp	yes	—	ISO-8601 timestamp	internal	—	—
source_arrival_timestamp	timestamp	yes	—	ISO-8601 timestamp	internal	—	—
sequence_number	bigint	yes	—	source-controlled / free text	internal	—	—

payment_gateway_status_event

10 July volume: 324,330 rows

Grain: immutable status-change event (CDC)

Primary key: status_event_id

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
status_event_id	varchar	yes	PK	source-controlled / free text	internal	—	—
gateway_txn_id	varchar	yes	FK → payment_gateway_log.gateway_txn_id	source-controlled / free text	internal	Replace with unresolved UNKNOWN-GATEWAY-LOG. **Rate:** 1.0% **Handling:** QUARANTINE_RECORD	UNKNOWN-GATEWAY-LOG
account_txn_id	bigint	no	FK → account_transaction.account_txn_id	source-controlled / free text	internal	—	—
card_txn_id	bigint	no	FK → card_transaction.card_txn_id	source-controlled / free text	internal	—	—
status	varchar	yes	—	source-controlled / free text	internal	—	—
status_timestamp	timestamp	yes	—	ISO-8601 timestamp	internal	—	—
source_arrival_timestamp	timestamp	yes	—	ISO-8601 timestamp	internal	—	—
sequence_number	bigint	yes	—	source-controlled / free text	internal	—	—

account_transaction

10 July volume: 575,674 rows

Grain: source entity / relationship snapshot (SCD2-ready)

Primary key: account_txn_id

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
account_txn_id	bigint	yes	PK	source-controlled / free text	internal	—	—
account_id	bigint	no	FK → account.account_id	source-controlled / free text	internal	Replace with unresolved −1. **Rate:** 1.0% **Handling:** QUARANTINE_RECORD	−1
customer_ref	varchar	no	FK → core_banking_customer.cust_no	source-controlled / free text	internal	—	—
amount	decimal(12,2)	no	—	numeric range	financial / risk	Set to −100.00 for otherwise valid CREDIT transactions. **Rate:** 1.0% of credits **Handling:** QUARANTINE_RECORD	−100.00
transaction_type	varchar	no	—	DEPOSIT / WITHDRAWAL / TRANSFER / FEE / INTEREST / REVERSAL / ACCOUNT_PAYMENT	internal	—	—
direction	varchar	no	—	DEBIT / CREDIT	internal	—	—
txn_timestamp	timestamp	no	—	ISO-8601 timestamp	internal	—	—
status	varchar	no	—	Latest source state only: INITIATED / PENDING / SUCCESS / REJECTED / CANCELED. Full state-change history is retained in account_transaction_status_event.	internal	Change clean SUCCESS to conflicting FAILED; event history remains intact. **Rate:** 1.0% of successful transactions **Handling:** RECONCILE_LATEST_STATUS	FAILED
channel	varchar	no	FK → transaction_channel.channel_id	source-controlled / free text	internal	—	—
merchant_id	varchar	no	FK → merchant.merchant_id	source-controlled / free text	internal	—	—

transaction_channel

10 July volume: 6 rows

Grain: source entity / relationship snapshot (SCD2-ready)

Primary key: channel_id

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
channel_id	varchar	yes	PK	source-controlled / free text	internal	—	—
channel_name	varchar	no	—	source-controlled / free text	internal	—	—
channel_type	varchar	no	—	source-controlled / free text	internal	—	—

merchant

10 July volume: 2,638 rows

Grain: source entity / relationship snapshot (SCD2-ready)

Primary key: merchant_id

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
merchant_id	varchar	yes	PK	source-controlled / free text	internal	—	—
merchant_name	varchar	no	—	source-controlled / free text	internal	—	—
mcc_code	varchar	no	—	source-controlled / free text	internal	—	—
country	varchar	no	—	source-controlled / free text	internal	—	—

merchant_store

10 July volume: 6,270 rows

Grain: source entity / relationship snapshot (SCD2-ready)

Primary key: store_id

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
store_id	varchar	yes	PK	source-controlled / free text	internal	—	—

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
merchant_id	varchar	no	FK → merchant.merchant_id	source-controlled / free text	internal	—	—
store_name	varchar	no	—	source-controlled / free text	internal	—	—
store_description	varchar	no	—	source-controlled / free text	internal	Set to NULL. **Rate:** 6.0% **Handling:** QUARANTINE_FIELD	NULL
store_type	varchar	no	—	source-controlled / free text	internal	—	—
store_address	varchar	no	—	source-controlled / free text	sensitive	—	—
risk_rating	varchar	no	—	source-controlled / free text	internal	Replace with UNKNOWN. **Rate:** 1.0% **Handling:** QUARANTINE_FIELD	UNKNOWN
registered_date	date	no	—	ISO date	internal	—	—

log_atm

10 July volume: 88,110 rows

Grain: source entity / relationship snapshot (SCD2-ready)

Primary key: log_id

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
log_id	varchar	yes	PK	source-controlled / free text	internal	—	—
card_id	varchar	no	FK → card.card_id	source-controlled / free text	internal	—	—
card_number	varchar	no	—	source-controlled / free text	sensitive	Replace with malformed INVALID-PAN. **Rate:** 1.0% **Handling:** QUARANTINE_FIELD	[malformed masked PAN]
account_txn_id	bigint	no	FK → account_transaction.account_txn_id	source-controlled / free text	internal	—	—
atm_id	varchar	no	—	source-controlled / free text	internal	—	—

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
txn_type	varchar	no	—	ATM action: WITHDRAWAL / BALANCE_INQUIRY / PIN_CHANGE / DEPOSIT. This is a raw ATM-switch event, not a card_transaction.	internal	—	—
amount	decimal(12,2)	no	—	numeric range	financial / risk	—	—
log_timestamp	timestamp	no	—	ISO-8601 timestamp	internal	—	—
response_code	varchar	no	—	source-controlled / free text	internal	Replace with unsupported UNKNOWN. **Rate:** 2.0% **Handling:** QUARANTINE_FIELD	UNKNOWN

payment_gateway_log

10 July volume: 108,110 rows

Grain: source entity / relationship snapshot (SCD2-ready)

Primary key: gateway_txn_id

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
gateway_txn_id	varchar	yes	PK	source-controlled / free text	internal	—	—
provider_payment_ref	varchar	no	—	source-controlled / free text	internal	—	—
payment_method	varchar	no	—	source-controlled / free text	internal	—	—
account_txn_id	bigint	no	FK → account_transaction.account_txn_id	source-controlled / free text	internal	—	—
card_txn_id	bigint	no	FK → card_transaction.card_txn_id	source-controlled / free text	internal	—	—
merchant_id	varchar	no	FK → merchant.merchant_id	source-controlled / free text	internal	—	—

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
amount	decimal(12,2)	no	—	numeric range	financial / risk	—	—
currency	varchar	no	—	source-controlled / free text	internal	Replace with invalid currency XXX. **Rate:** 0.5% **Handling:** QUARANTINE_FIELD	XXX
gateway_status	varchar	no	—	source-controlled / free text	internal	—	—
gateway_timestamp	timestamp	no	—	ISO-8601 timestamp	internal	—	—
provider_code	varchar	no	—	source-controlled / free text	internal	—	—

balance_snapshot

10 July volume: 127,207 rows

Grain: source entity / relationship snapshot (SCD2-ready)

Primary key: balance_id

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
balance_id	varchar	yes	PK	source-controlled / free text	internal	—	—
account_id	bigint	no	FK → account.account_id	source-controlled / free text	internal	—	—
balance_date	date	no	—	ISO date	internal	—	—
opening_balance	decimal(14,2)	no	—	numeric range	financial / risk	—	—
closing_balance	decimal(14,2)	no	—	numeric range	financial / risk	Set to −250.00 only for snapshots whose account product is SAVINGS. **Rate:** 0.5% of eligible savings rows **Handling:** QUARANTINE_RECORD	−250.00
available_balance	decimal(14,2)	no	—	numeric range	financial / risk	—	—

Card

card_transaction_status_event

10 July volume: 1,152,972 rows

Grain: immutable status-change event (CDC)

Primary key: status_event_id

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
status_event_id	varchar	yes	PK	source-controlled / free text	internal	—	—
card_txn_id	bigint	yes	FK → card_transaction.card_txn_id	source-controlled / free text	internal	Replace with unresolved −1. **Rate:** 1.0% **Handling:** QUARANTINE_RECORD	−1
status	varchar	yes	—	source-controlled / free text	internal	—	—
status_timestamp	timestamp	yes	—	ISO-8601 timestamp	internal	—	—
source_arrival_timestamp	timestamp	yes	—	ISO-8601 timestamp	internal	—	—
sequence_number	bigint	yes	—	source-controlled / free text	internal	—	—

card

10 July volume: 77,026 rows

Grain: source entity / relationship snapshot (SCD2-ready)

Primary key: card_id

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
card_id	varchar	yes	PK	source-controlled / free text	internal	—	—
account_id	bigint	no	FK → account.account_id	source-controlled / free text	internal	Replace with unresolved −1. **Rate:** 1.0% **Handling:** QUARANTINE_RECORD	−1

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
card_number	varchar	no	—	source-controlled / free text	sensitive	Replace with malformed INVALID-PAN. **Rate:** 1.0% **Handling:** QUARANTINE_FIELD	[malformed masked PAN]
card_type	varchar	no	—	source-controlled / free text	internal	—	—
issue_date	date	no	—	ISO date	internal	—	—
expiry_date	date	no	—	ISO date	internal	—	—
status	varchar	no	—	source-controlled / free text	internal	—	—

card_transaction

10 July volume: 384,324 rows

Grain: source entity / relationship snapshot (SCD2-ready)

Primary key: card_txn_id

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
card_txn_id	bigint	yes	PK	source-controlled / free text	internal	—	—
card_id	varchar	no	FK → card.card_id	source-controlled / free text	internal	Replace with unresolved UNKNOWN-CARD. **Rate:** 1.0% **Handling:** QUARANTINE_RECORD	UNKNOWN-CARD
merchant_id	varchar	no	FK → merchant.merchant_id	source-controlled / free text	internal	—	—
amount	decimal(12,2)	no	—	numeric range	financial / risk	—	—
card_transaction_type	varchar	no	—	PURCHASE / REFUND / REVERSAL / FEE. ATM withdrawals are excluded and belong to account_transaction plus log_atm.	internal	—	—
txn_timestamp	timestamp	no	—	ISO-8601 timestamp	internal	—	—
txn_status	varchar	no	—	source-controlled / free text	internal	—	—

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
is_fraud	boolean	no	—	source-controlled / free text	internal	—	—

card_fraud_flag

10 July volume: 8,454 rows

Grain: source entity / relationship snapshot (SCD2-ready)

Primary key: flag_id

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
flag_id	varchar	yes	PK	source-controlled / free text	internal	—	—
card_txn_id	bigint	no	FK → card_transaction. card_txn_id	source-controlled / free text	internal	Replace with unresolved −1. **Rate:** 1.0% **Handling:** QUARANTINE_RECORD	−1
flag_reason	varchar	no	—	source-controlled / free text	internal	—	—
flag_date	date	no	—	ISO date	internal	—	—
resolved_status	varchar	no	—	source-controlled / free text	internal	—	—

card_limit_history

10 July volume: 37,106 rows

Grain: source entity / relationship snapshot (SCD2-ready)

Primary key: history_id

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
history_id	varchar	yes	PK	source-controlled / free text	internal	—	—
card_id	varchar	no	FK → card.card_id	source-controlled / free text	internal	Replace with unresolved UNKNOWN-CARD. **Rate:** 1.0% **Handling:** QUARANTINE_RECORD	UNKNOWN-CARD

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
limit_amount	decimal(12,2)	no	—	numeric range	financial / risk	Physical type changes from decimal(12,2) to parseable decimal string for the 2026-07-08 snapshot. **Rate:** one snapshot **Handling:** CAST_TO_CONTRACT_TYPE	[controlled injected value]
effective_date	date	no	—	ISO date	internal	—	—

Financial Crime

fraud_alert

10 July volume: 17,268 rows

Grain: source entity / relationship snapshot (SCD2-ready)

Primary key: alert_id

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
alert_id	varchar	yes	PK	source-controlled / free text	internal	—	—
account_txn_id	bigint	no	FK → account_transaction.account_txn_id	source-controlled / free text	internal	Replace with unresolved −1. **Rate:** 1.0% **Handling:** QUARANTINE_RECORD	−1
alert_type	varchar	no	—	source-controlled / free text	internal	—	—
alert_score	decimal(5,2)	no	—	numeric range	financial / risk	Set to 101.00, above the valid 0–100 range. **Rate:** 1.0% **Handling:** QUARANTINE_RECORD	101.00
alert_status	varchar	no	—	source-controlled / free text	internal	—	—
created_date	date	no	—	ISO date	internal	—	—

transaction_monitoring_alert

10 July volume: 11,016 rows

Grain: source entity / relationship snapshot (SCD2-ready)

Primary key: alert_id

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
alert_id	varchar	yes	PK	source-controlled / free text	internal	—	—
customer_ref	varchar	no	FK → core_banking_customer.cust_no	source-controlled / free text	internal	—	—
primary_account_txn_id	bigint	no	—	source-controlled / free text	internal	—	—
alert_type	varchar	no	—	source-controlled / free text	internal	—	—
alert_score	decimal(5,2)	no	—	numeric range	financial / risk	Set to 101.00, above the valid 0–100 range. **Rate:** 1.0% **Handling:** QUARANTINE_RECORD	101.00
alert_status	varchar	no	—	source-controlled / free text	internal	Change to CLOSED where the related investigation link makes that status inconsistent. **Rate:** 1.0% of eligible linked alerts **Handling:** RECONCILE_LATEST_STATUS	CLOSED
alert_timestamp	timestamp	no	—	ISO-8601 timestamp	internal	—	—

transaction_monitoring_alert_account_transaction

10 July volume: 11,016 rows

Grain: source entity / relationship snapshot (SCD2-ready)

Primary key: alert_account_txn_link_id

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
alert_account_txn_link_id	varchar	yes	PK	source-controlled / free text	internal	—	—
alert_id	varchar	no	FK → transaction_monitoring_alert.alert_id	source-controlled / free text	internal	—	—
account_txn_id	bigint	no	FK → account_transaction.account_txn_id	source-controlled / free text	internal	Replace with unresolved −1. **Rate:** 1.0% **Handling:** QUARANTINE_RECORD	−1

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
is_primary	boolean	no	—	source-controlled / free text	internal	—	—

transaction_monitoring_alert_card_transaction

10 July volume: 3,750 rows

Grain: source entity / relationship snapshot (SCD2-ready)

Primary key: alert_card_txn_link_id

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
alert_card_txn_link_id	varchar	yes	PK	source-controlled / free text	internal	—	—
alert_id	varchar	no	FK → transaction_monitoring_alert.alert_id	source-controlled / free text	internal	—	—
card_txn_id	bigint	no	FK → card_transaction.card_txn_id	source-controlled / free text	internal	Replace with unresolved −1. **Rate:** 1.0% **Handling:** QUARANTINE_RECORD	−1
is_primary	boolean	no	—	source-controlled / free text	internal	—	—

investigation_case_transaction_monitoring_alert

10 July volume: 5,016 rows

Grain: source entity / relationship snapshot (SCD2-ready)

Primary key: case_alert_link_id

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
case_alert_link_id	varchar	yes	PK	source-controlled / free text	internal	—	—
case_id	varchar	no	FK → investigation_case.case_id	source-controlled / free text	internal	—	—

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
alert_id	varchar	no	FK → transaction_monitoring_alert.alert_id	source-controlled / free text	internal	Replace with unresolved UNKNOWN-MONITORING-ALERT. **Rate:** 1.0% **Handling:** QUARANTINE_RECORD	UNKNOWN-MONITORING-ALERT
linked_timestamp	timestamp	no	—	ISO-8601 timestamp	internal	—	—

investigation_case_card_fraud_flag

10 July volume: 6,600 rows

Grain: source entity / relationship snapshot (SCD2-ready)

Primary key: case_flag_link_id

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
case_flag_link_id	varchar	yes	PK	source-controlled / free text	internal	—	—
case_id	varchar	no	FK → investigation_case.case_id	source-controlled / free text	internal	—	—
flag_id	varchar	no	FK → card_fraud_flag.flag_id	source-controlled / free text	internal	—	—
linked_timestamp	timestamp	no	—	ISO-8601 timestamp	internal	—	—

investigation_case

10 July volume: 13,516 rows

Grain: source entity / relationship snapshot (SCD2-ready)

Primary key: case_id

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
case_id	varchar	yes	PK	source-controlled / free text	internal	—	—
investigation_type	varchar	no	—	FRAUD / AML	internal	—	—
case_origin	varchar	no	—	ALERT / CUSTOMER_DISPUTE / CHARGEBACK / SANCTIONS_HIT / MANUAL	internal	—	—
case_status	varchar	no	—	OPEN / IN_REVIEW / ESCALATED / CLOSED	internal	—	—
priority	varchar	no	—	LOW / MEDIUM / HIGH / CRITICAL	internal	—	—
opened_timestamp	timestamp	no	—	ISO-8601 timestamp	internal	—	—
closed_timestamp	timestamp	no	—	ISO-8601 timestamp	internal	—	—
assigned_analyst_id	varchar	no	—	source-controlled / free text	internal	—	—

investigation_case_account_transaction

10 July volume: 6,000 rows

Grain: source entity / relationship snapshot (SCD2-ready)

Primary key: case_account_txn_link_id

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
case_account_txn_link_id	varchar	yes	PK	source-controlled / free text	internal	—	—
case_id	varchar	no	FK → investigation_case.case_id	source-controlled / free text	internal	—	—
account_txn_id	bigint	no	FK → account_transaction.account_txn_id	source-controlled / free text	internal	unresolved transaction reference; 1.0%	[unresolved synthetic reference]

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
link_reason	varchar	no	—	PRIMARY_TRANSACTION / RELATED_TRANSACTION / CUSTOMER_DISPUTE	internal	—	—
linked_timestamp	timestamp	no	—	ISO-8601 timestamp	internal	—	—

investigation_case_card_transaction

10 July volume: 3,000 rows

Grain: source entity / relationship snapshot (SCD2-ready)

Primary key: case_card_txn_link_id

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
case_card_txn_link_id	varchar	yes	PK	source-controlled / free text	internal	—	—
case_id	varchar	no	FK → investigation_case.case_id	source-controlled / free text	internal	—	—
card_txn_id	bigint	no	FK → card_transaction.card_txn_id	source-controlled / free text	internal	unresolved transaction reference; 1.0%	[unresolved synthetic reference]
link_reason	varchar	no	—	PRIMARY_TRANSACTION / RELATED_TRANSACTION / CUSTOMER_DISPUTE	internal	—	—
linked_timestamp	timestamp	no	—	ISO-8601 timestamp	internal	—	—

investigation_case_fraud_alert

10 July volume: 6,000 rows

Grain: source entity / relationship snapshot (SCD2-ready)

Primary key: case_alert_link_id

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
case_alert_link_id	varchar	yes	PK	source-controlled / free text	internal	—	—
case_id	varchar	no	FK → investigation_case.case_id	source-controlled / free text	internal	—	—
alert_id	varchar	no	FK → fraud_alert.alert_id	source-controlled / free text	internal	unresolved alert reference; 1.0%	[unresolved synthetic reference]
linked_timestamp	timestamp	no	—	ISO-8601 timestamp	internal	—	—

investigation_case_sanction_screening

10 July volume: 2,000 rows

Grain: source entity / relationship snapshot (SCD2-ready)

Primary key: case_screening_link_id

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
case_screening_link_id	varchar	yes	PK	source-controlled / free text	internal	—	—
case_id	varchar	no	FK → investigation_case.case_id	source-controlled / free text	internal	—	—
screening_id	varchar	no	FK → sanction_screening.screening_id	source-controlled / free text	internal	unresolved screening reference; 1.0%	[unresolved synthetic reference]
linked_timestamp	timestamp	no	—	ISO-8601 timestamp	internal	—	—

investigation_note

10 July volume: 23,516 rows

Grain: source entity / relationship snapshot (SCD2-ready)

Primary key: note_id

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
note_id	varchar	yes	PK	source-controlled / free text	internal	—	—
case_id	varchar	no	FK → investigation_case.case_id	source-controlled / free text	internal	—	—
author_id	varchar	no	—	source-controlled / free text	internal	—	—
note_timestamp	timestamp	no	—	ISO-8601 timestamp	internal	—	—
source_arrival_timestamp	timestamp	yes	—	ISO-8601 timestamp	internal	Shift 600 seconds later; the note's business time is unchanged. **Rate:** 1.0% **Handling:** PROCESS_BY_ARRIVAL_ORDER Physical type changes from timestamp to ISO-8601 string for the 2026-07-06 snapshot. **Rate:** one snapshot **Handling:** CAST_TO_CONTRACT_TYPE	source-arrival timestamp +600 seconds [controlled injected value]
note_type	varchar	no	—	ANALYST_NOTE / EVIDENCE / CUSTOMER_CONTACT / DECISION	internal	—	—
note_text	varchar	no	—	source-controlled / free text	restricted content	—	—

aml_case

10 July volume: 8,266 rows

Grain: source entity / relationship snapshot (SCD2-ready)

Primary key: case_id

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
case_id	varchar	yes	PK	source-controlled / free text	internal	—	—
investigation_case_id	varchar	no	Unique	source-controlled / free text	internal	Replace with unresolved UNKNOWN-CASE. **Rate:** 1.0% **Handling:** QUARANTINE_RECORD	UNKNOWN-CASE

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
customer_ref	varchar	no	—	source-controlled / free text	internal	—	—
case_type	varchar	no	—	source-controlled / free text	internal	—	—
risk_level	varchar	no	—	source-controlled / free text	internal	—	—
opened_date	date	no	—	ISO date	internal	—	—
closed_date	date	no	—	ISO date	internal	—	—

sanction_screening

10 July volume: 23,516 rows

Grain: source entity / relationship snapshot (SCD2-ready)

Primary key: screening_id

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
screening_id	varchar	yes	PK	source-controlled / free text	internal	—	—
customer_ref	varchar	no	—	source-controlled / free text	internal	—	—
watchlist_id	varchar	no	—	source-controlled / free text	internal	—	—
screened_name	varchar	no	—	source-controlled / free text	direct identifier	Truncate the name. **Rate:** 2.0% **Handling:** QUARANTINE_FIELD	[truncated synthetic name]
match_score	decimal(5,2)	no	—	numeric range	internal	—	—
screening_date	date	no	—	ISO date	internal	—	—
result	varchar	no	—	source-controlled / free text	internal	Change eligible HIT results to conflicting CLEAR. **Rate:** 1.0% of HIT rows **Handling:** RECONCILE_LATEST_STATUS	CLEAR

suspicious_activity_report

10 July volume: 1,710 rows

Grain: source entity / relationship snapshot (SCD2-ready)

Primary key: sar_id

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
sar_id	varchar	yes	PK	source-controlled / free text	internal	—	—
case_id	varchar	no	FK → aml_case.case_id	source-controlled / free text	internal	Replace with unresolved UNKNOWN-AML-CASE. **Rate:** 1.0% **Handling:** QUARANTINE_RECORD	UNKNOWN-AML-CASE
filed_date	date	no	—	ISO date	internal	—	—
regulatory_ref	varchar	no	—	source-controlled / free text	internal	—	—
status	varchar	no	—	source-controlled / free text	internal	—	—

watchlist

10 July volume: 2,638 rows

Grain: source entity / relationship snapshot (SCD2-ready)

Primary key: watchlist_id

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
watchlist_id	varchar	yes	PK	source-controlled / free text	internal	—	—
entity_name	varchar	no	—	source-controlled / free text	internal	—	—
list_type	varchar	no	—	source-controlled / free text	internal	—	—
country	varchar	no	—	source-controlled / free text	internal	—	—
added_date	date	no	—	ISO date	internal	Shift 3,650 days into the past. **Rate:** 1.0% **Handling:** FLAG_STALE	source date − 3,650 days

account_transaction_risk_score

10 July volume: 575,674 rows

Grain: source entity / relationship snapshot (SCD2-ready)

Primary key: score_id

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
score_id	varchar	yes	PK	source-controlled / free text	internal	—	—
account_txn_id	bigint	no	Unique	source-controlled / free text	internal	Replace with unresolved −1. **Rate:** 1.0% **Handling:** QUARANTINE_RECORD	−1
model_score	decimal(6,4)	no	—	numeric range	financial / risk	—	—
risk_band	varchar	no	—	source-controlled / free text	internal	—	—
scored_date	date	no	—	ISO date	internal	—	—

call_center_log

10 July volume: 54,052 rows

Grain: source entity / relationship snapshot (SCD2-ready)

Primary key: call_id

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
call_id	varchar	yes	PK	source-controlled / free text	internal	—	—
customer_ref	varchar	no	—	source-controlled / free text	internal	—	—
caller_phone	varchar	no	—	source-controlled / free text	contact	—	—
call_timestamp	timestamp	no	—	ISO-8601 timestamp	internal	—	—
call_reason	varchar	no	—	source-controlled / free text	internal	—	—
agent_id	varchar	no	—	source-controlled / free text	internal	—	—

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
call_duration_seconds	bigint	no	—	source-controlled / free text	internal	Physical type changes from bigint to parseable decimal string for the 2026-07-05 snapshot. **Rate:** one snapshot **Handling:** CAST_TO_CONTRACT_TYPE	[controlled injected value]
case_id	varchar	no	FK → investigation_case.case_id.	source-controlled / free text	internal	—	—

chargeback

10 July volume: 5,404 rows

Grain: source entity / relationship snapshot (SCD2-ready)

Primary key: chargeback_id

Field	Type	Required	Key / reference	Accepted values	Classification	Error injected	Injected example
chargeback_id	varchar	yes	PK	source-controlled / free text	internal	—	—
card_txn_id	bigint	no	FK → card_transaction.card_txn_id	source-controlled / free text	internal	—	—
reason_code	varchar	no	—	source-controlled / free text	internal	Replace with unsupported UNKNOWN. **Rate:** 1.0% **Handling:** QUARANTINE_FIELD	UNKNOWN
dispute_amount	decimal(12,2)	no	—	numeric range	internal	—	—
filed_date	date	no	—	ISO date	internal	—	—
status	varchar	no	—	source-controlled / free text	internal	—	—
resolved_date	date	no	—	ISO date	internal	—	—

Schema change timeline

Schema changes are deliberate source-contract evolution, not data-quality defects. Each change is applied to carried-forward clean state before that date's inserts and complete after-images are produced. Consumers must apply the dated contract before comparing fields between snapshots.

Effective date	Source table	Field change	Effect on carried-forward rows	Required ingestion behaviour
2026-07-05	payment_gateway_log	gateway_provider renamed to provider_code	Values are retained under the new field name.	Explicit column mapping
2026-07-06	crm_customer	Add preferred_contact_method (string)	Existing rows receive deterministic weighted values: EMAIL, SMS, or PHONE (60/25/15).	Evolve schema
2026-07-07	account_transaction_risk_score	model_score changes from decimal(5,4) to decimal(6,4)	Existing values are strictly cast; no intended business-value change.	Safe contract widening
2026-07-07	log_atm	Retire location	The field is absent from that delivery and later cumulative state.	Retire column
2026-07-09	account	Add servicing_model (string)	Existing rows receive deterministic weighted values: DIGITAL, BRANCH_ASSISTED, or RELATIONSHIP_MANAGED (60/30/10).	Evolve schema
2026-07-10	payment_gateway_log	payment_ref renamed to provider_payment_ref	Values are retained under the new field name.	Explicit column mapping

Snapshot physical-type test cases

These are deliberately malformed output copies for ingestion testing. They do not change clean cumulative state and are not normal SCD2 updates.

Business date	Field	Contract type → injected physical type	Recovery date	Expected handling
2026-07-05	call_center_log.call_duration_seconds	bigint → parseable decimal-text string	2026-07-06	CAST_TO_CONTRACT_TYPE
2026-07-06	investigation_note.source_arrival_timestamp	timestamp → ISO-8601 string	2026-07-07	CAST_TO_CONTRACT_TYPE
2026-07-08	card_limit_history.limit_amount	decimal(12,2) → parseable decimal-text string	2026-07-09	CAST_TO_CONTRACT_TYPE

Technical implementation

The mock-data set is configuration-driven and deterministic. The implementation separates clean source generation, legitimate source changes, deliberate defects, snapshot reconstruction, quality audit, and publication so that a failed scenario can be reproduced without turning intentional errors into undocumented behaviour.

Implementation stage	Technical approach
Source contracts	Define source-native table schema, primary key, references, data types, and allowed source values for all 41 tables. Mixed customer references remain raw-source attributes; identity conformance is deferred downstream.
Deterministic generation	Use a fixed run configuration and seed. Generate valid Customer Master records first, then transaction/card entities, then Financial Crime relationships so the clean parent graph resolves before defects are applied.
Daily changes	Produce 259,070 new clean keys and 1,321 complete after-images per business date. Sixteen mutable tables emit changed full rows with the same primary key; other facts, logs, histories, notes, evidence, and status events append.
Error injection	Apply the documented, deterministic defects only after the valid base and legitimate source changes. Store affected-key evidence and expected handling in a local-only manifest; never publish it as Bronze data.
Snapshot reconstruction	Maintain clean cumulative state by business date. Use the latest complete after-image for mutable entities; accumulate append-only facts and events. Apply physical-type test errors only to output copies, never to clean state.
Validation and audit	Validate row counts, schema state, primary-key/relationship expectations, controlled error coverage, and 41-table snapshot completeness before publication.
Publication	Publish audited, full Parquet snapshots only. Raw daily inputs, manifests, cache/state, change reports, and quality reports remain outside the published hierarchy.

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```
flowchart LR
    A[Source contracts and fixed seed] --> B[Valid multi-domain base generation]
    B --> C[Legitimate daily inserts and complete after-images]
    C --> D[Apply effective-date schema changes]
    D --> E[Apply deterministic error injections]
    E --> F[Local-only truth manifest]
    E --> G[Clean cumulative state and snapshot reconstruction]
    G --> H[41-table quality audit]
    H --> I[Audited Parquet full snapshots]
    I --> J[Published S3 source hierarchy]
    F -. restricted; never published .-> K[Evidence boundary]
```

Running the generator and configuring scenarios

Prerequisites

The implementation runs on Python 3.11+ with Java 17 recommended for Spark. Create the project environment once before running the v2 workflow:

BASH

```
make setup
```

Standard v2 generation sequence

Run the commands from the repository root, in this order:

BASH

```bash
# 1. Generate the clean 4 July seed and daily source changes for 5–10 July
make banking-final-local

# 2. Reconstruct the six complete point-in-time Parquet snapshots
make banking-snapshots-local

# 3. Validate snapshot counts, schema state, and publication readiness
make banking-snapshots-quality

# 4. Publish only the audited full snapshots to the configured S3 destination
make banking-snapshots-s3
```

The first command runs the four domain generators in dependency order, then generates the daily stateful changes and validates the resulting source simulation. Snapshot build state supports a matching interrupted run being resumed; the quality gate must pass before the upload command runs.

YAML configuration layers

Configuration layer	YAML setting family	What it controls	Change guidance
Domain entry points	customer_master_production/run.yaml, customer_transaction_production/run.yaml, card_production/run.yaml, financial_crime_production/run.yaml	Domain tables, seeds, dependencies, outputs, and the linked error scenario.	Use when changing the size, scope, or dependency order of a source domain.
Table contracts	One YAML file per source table, such as account.yaml, card.yaml, or investigation_case.yaml	Fields, source-native types, keys, parent references, generators, distributions, and row counts.	Change only with a matching schema/data-dictionary update and contract test.
Error scenarios	error_injecting.yaml for each domain	Deterministic field/row defects, selection rate, exclusions, expected handling, and local-only evidence manifest.	Change to add or tune a quality scenario; keep clean and injected output generation paired.
Daily delivery	daily/run.yaml plus per-domain daily settings	Business-date processing, state location, daily inserts, complete after-images, append-only behaviour, and replay scenarios.	Change to alter daily volume, mutable-table behaviour, or the delivery date range.
Schema evolution	snapshots/schema_timeline.yaml	Effective-dated add, rename, widen, and retire actions.	Add dated contract changes here; distinguish source evolution from defects.
Snapshot type tests	snapshots/schema_error_injecting.yaml	Output-only physical-type test cases and recovery dates.	Use only for parseable ingestion-error scenarios; do not apply to clean cumulative state.
Snapshot publication	snapshots/run.yaml	Snapshot dates, full-snapshot layout, audit expectations, and publication inputs.	Change only when the delivery calendar or publishing contract changes.

Safe configuration workflow

1. Update the appropriate YAML contract, scenario, or timeline file.
2. Run make test-v2 to confirm configuration and generator contracts.
3. Generate the v2 source workflow locally and run the snapshot quality gate.
4. Review the 41-table audit results and the documented error coverage.
5. Publish only after the S3 upload guard accepts the audited full snapshots.

Testing and verification

The generator is tested as a source-data system, not only as a collection of record generators. Automated tests and operational quality gates verify that the contracts, daily behaviour, deliberate defects, snapshots, and publication controls work together.

Test layer	What is verified	Expected result
Configuration and schema contracts	The four domain configurations, 41-table DBML contract, table keys, data types, references, ranges, and source-model layout.	The active source model is internally consistent before data is generated.
Generator primitives	Generic, range, transaction, YAML, and custom generator behaviour, using a bounded fixture where a small dependency graph is sufficient.	Generated values follow their configured type, range, and source-format contract.
Valid-base relationships	Customer, account, card, transaction, alert, case, AML, and evidence paths are created in dependency order.	Clean parent/child relationships resolve before deliberate defects are applied.
Daily engine	Stateful daily generation, deterministic row selection, complete after-images, append-only facts/events, disjoint updates, and rejection of clean no-op updates.	Each business date produces the expected inserts and genuine changes without silently replaying unchanged rows.
Error-injection scenarios	Scenario parsing, deterministic rule application, changed-row evidence, expected handling, and local-only manifest behaviour.	Every documented defect is reproducible and its intended downstream response is testable.
Snapshot construction	Point-in-time reconstruction, resumable state, row totals, schema timeline application, and output-only physical-type error cases.	Each full date is complete, carries the correct source contract, and preserves clean cumulative state.
S3 publication guards	Snapshot inventory, allowable dates, Parquet-only payloads, required 41-table quality report, destination controls, and file/byte parity.	Only audited full snapshots are uploaded; samples, raw deliveries, manifests, and incomplete publications are rejected.

Test execution

Run the retained v2 automated suite from the repository root:

BASH

```bash
make test-v2
```

The suite executes the v2 tests with the project virtual environment. Production quality-gate commands are separate operational validators; they check the generated snapshot artifacts and publication prerequisites in addition to unit and contract tests.

SCD Type 2 and CDC design

SCD Type 2: every non-event table

All non-event tables are **SCD2-ready**. The source delivery publishes complete daily after-images; a downstream historical layer can therefore version each entity or relationship without overwriting prior state.

SCD2 component	Design in this mock-data model
Business key	Use the source primary key for the table (for example, the customer, account, card, merchant, case, or bridge-link ID).
Change detection	Compare a canonical row hash of non-technical source attributes between successive full snapshots.
Version start	Set effective_from to the first business date on which the new after-image appears.
Version end	Set effective_to to the business date immediately before the next changed version; leave open for the current version.
Current marker	Set is_current = true only for the latest open version of each business key.
Version number	Increment version_number per business key when the row hash changes.
Delete / absence policy	Interpret an absent source row as a configured source lifecycle condition, not an automatic hard delete. Close the SCD2 row only when the delivery contract confirms deletion/inactivation semantics.
Relationship history	Apply the same approach to bridge tables such as customer-account and case-evidence links so ownership and evidence associations are historically reproducible.

Non-event examples include customer, account, KYC, employment, request, merchant, gateway log, balance snapshot, card, fraud flag, risk score, case, AML, screening, watchlist, and evidence-bridge tables. A balance snapshot remains an effective-dated source fact; SCD2 preserves changes to a delivered row while its balance_date expresses the source observation date.

CDC: immutable status-event tables

The following tables are native immutable event feeds rather than SCD2 dimensions:

Event table	Event identity	Parent business key	CDC order	Late/replay handling
account_transaction_status_event	status_event_id	account_txn_id	status_timestamp, then sequence_number	Deduplicate by event ID; preserve late arrival and process arrival order where required
card_transaction_status_event	status_event_id	card_txn_id	status_timestamp, then sequence_number	Deduplicate by event ID; do not overwrite history

Event table	Event identity	Parent business key	CDC order	Late/replay handling
atm_transaction_status_event	status_event_id	log_id	status_timestamp, then sequence_number	Deduplicate by event ID; retain optional transaction reference as structural null where valid
payment_gateway_status_event	status_event_id	gateway_txn_id	status_timestamp, then sequence_number	Deduplicate by event ID; retain account/card polymorphic context

CDC rule: retain every accepted immutable event. Use the event ID for idempotency, use business time plus source sequence to order the event stream, and retain source-arrival time to test late-arrival handling. A replay is deliberately injected as a CDC deduplication test, not as an SCD2 version.

MERMAID

```
flowchart LR
    S[Daily full source snapshot] --> K[Identify table type]
    K -->|Non-event table| H[Compare by business key and row hash]
    H -->|New or changed| V[Close prior version and create SCD2 version]
    H -->|Unchanged| C[Keep current SCD2 version]
    K -->|Status-event table| E[Read immutable event]
    E --> D[Deduplicate by event ID]
    D --> O[Order by business timestamp + sequence]
    O --> L[Retain late arrival metadata]
```

Daily snapshots and controlled S3 delivery

Business date	Full-snapshot rows
5 July 2026	4,941,678
6 July 2026	5,201,626
7 July 2026	5,461,574
8 July 2026	5,721,522
9 July 2026	5,981,470
10 July 2026	6,241,418

Each clean daily delivery contains **259,070 inserts**, **1,321 genuine complete after-image updates**, and **878 injected account-status replay rows** for deduplication testing. The series also contains controlled source evolution: added attributes, renamed fields, widened numeric ranges, retired attributes, and parseable physical-type errors before a clean final schema.

TEXT

```
s3://nab-src-dataset/banking_v2/snapshots/
  simulation_id=banking-20260705-20260710/
```

```
snapshot_type=full/business_date=2026-07-05/<domain>/<table>/
snapshot_type=full/business_date=2026-07-06/<domain>/<table>/
snapshot_type=full/business_date=2026-07-07/<domain>/<table>/
snapshot_type=full/business_date=2026-07-08/<domain>/<table>/
snapshot_type=full/business_date=2026-07-09/<domain>/<table>/
snapshot_type=full/business_date=2026-07-10/<domain>/<table>/
```

Only the six expected full dates, valid Parquet payloads, and a successful 41-table quality report are eligible for publication. Raw daily deliveries, truth manifests, build/cache state, change reports, and quality reports remain local. File and byte parity are checked after upload.

AI access and acceptance

AI access rule: Raw Bronze source tables and the error-injection truth manifest must not be exposed directly to AI consumers. Any AI use case must use an approved, minimum-necessary downstream view with masking, tokenisation, redaction, or removal.

- **Coverage:** Every full snapshot includes all 41 published tables.
- **Relationships:** Clean base rows resolve before intentional defects are applied; references are tested across customer, transaction, card, alert, case, and AML paths.
- **Quality:** Required error categories are represented and valid structural nulls are not counted as defects.
- **History:** Non-event tables can be historised as SCD2; immutable status-event tables use idempotent CDC.
- **Delivery:** Only audited full Parquet snapshots enter the S3 simulation.
- **Safety:** Raw restricted data and truth labels are blocked from direct AI access.